



# Influence of source structure and background wind on the excitation, propagation, and structure of Lamb and Pekeris waves

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## Abstract

The excitation, propagation, and structure of eruption-induced Lamb and Pekeris waves, as well as the influence of background winds on them, are investigated using a linear hydrostatic atmospheric model. The amplitudes of the excited waves and the time required to transition to the theoretical structures predicted by classical tidal theory vary substantially depending on the horizontal extent and altitude of the initial temperature perturbation representing the eruption. Numerical experiments also demonstrate that the phase speed, node altitude, and damping characteristics of the Pekeris wave can deviate significantly from their theoretical counterparts in the presence of background winds; the underlying physical mechanisms responsible for these modifications are also examined. Furthermore, satellite-based visualization of the Lamb and Pekeris waves excited by the eruption of the Hunga Tonga-Hunga Ha'apai volcano reveals a clear difference in the damping characteristics between the eastward- and westward-propagating Pekeris waves, which is consistent with the results of our numerical calculations. These wind effects on the Pekeris wave are also consistent with recent simulation results using a high-top general circulation model.

**Keywords** Lamb wave · Pekeris wave · Initial value problem · Hunga Tonga-Hunga Ha'apai eruption

## 1 Introduction

The eruption of the Hunga Tonga-Hunga Ha'apai (HTHH) volcano is one of the most powerful and significant volcanic events since the beginning of satellite observations. The main eruption of HTHH occurred shortly after 04:00 UTC on January 15, 2022, and its volcanic plume reached the mesosphere for the first time in recorded history (e.g., Carr et al. 2022; Proud et al. 2022). This large-scale eruption also generated oscillations in various media, such as atmospheric waves (e.g., Wright et al. 2022; Matoza et al. 2022), ionospheric fluctuations (e.g., Themens et al. 2022; Wright et al. 2022), tsunamis (e.g., Carvajal et al. 2022; Kubota et al. 2022), and seismic ground motions (e.g., Matoza et al. 2022), which were detected worldwide.

Among the various types of atmospheric waves excited by the HTHH eruption, Lamb waves (Lamb 1911) were observed to propagate around the globe. Wright et al. (2022) analyzed brightness temperature data from multiple satellites covering various spectral bands and detected Lamb waves over a wide range of altitudes. In addition, Rose et al. (2024) detected Lamb waves through both satellite observations and surface pressure measurements from nationwide barometer networks in Japan, India, and the United States. Furthermore, Otsuka (2022) clearly visualized the wavefronts of Lamb waves by applying a second-order time difference to brightness temperatures observed in Band 8 of the geostationary satellite Himawari-8, which is primarily sensitive to radiation from the troposphere.

Recent satellite observations have also enabled the first detection of Pekeris waves. Watanabe et al. (2022) visualized the atmospheric waves excited by the HTHH eruption by calculating first-order time differences of brightness temperatures observed in Band 12 of Himawari-8, which is primarily sensitive to ozone-related radiation from the stratosphere. Among these waves, the fastest-propagating wavefront is the aforementioned Lamb wave, followed by the distinct Pekeris wave, which is later visualized in

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Sect. 2. Based on the azimuthally averaged propagation distances of their wavefronts approximately three hours after the HTHH eruption, Watanabe et al. (2022) showed that the phase speed of the Lamb wave is about  $315 \text{ m s}^{-1}$ , while that of the Pekeris wave is about  $245 \text{ m s}^{-1}$ . These phase speeds correspond to equivalent depths of 10.1 km for the Lamb wave and 6.1 km for the Pekeris wave.

In addition to their analysis based on satellite observations, Watanabe et al. (2022) detected signals of the Lamb and Pekeris waves in barograms recorded at more than 1500 sites across Japan. Although the amplitudes of the Lamb and Pekeris waves visualized by satellite observations were comparable, the surface pressure fluctuations associated with the Pekeris wave were an order of magnitude smaller than those of the Lamb wave.

Watanabe et al. (2022) also conducted simulations of Lamb and Pekeris waves excited by the volcanic eruption using JAGUAR (Watanabe and Miyahara 2009), a hydrostatic atmospheric general circulation model (AGCM) covering the entire neutral atmosphere. The initial temperature perturbation was imposed between the 800 hPa and 400 hPa levels, with a radius of 140 km, and its amplitude increased linearly with height from 50 K to 100 K. This simulation was nudged toward the background atmospheric state at the time of the HTHH eruption based on atmospheric reanalysis data. As a result, both Lamb and Pekeris waves were excited and propagated with phase speeds generally consistent with satellite observations. Under this experimental setup, the amplitudes of the sea level pressure (SLP) anomalies associated with the Lamb and Pekeris waves were comparable to those observed at ground-based stations. Furthermore, Watanabe et al. (2022) presented the vertical structures of the vertical pressure velocity ( $p$ -velocity) at the western edge of the wavefronts of the Lamb and Pekeris waves approximately four hours after the HTHH eruption. The  $p$ -velocity associated with the Lamb wave was vertically in phase, while that associated with the Pekeris wave exhibited a single node near the 90 hPa level.

Suzuki et al. (2023) also performed simulations of Lamb and Pekeris waves to investigate their respective contributions to sea-level fluctuations observed in Japan, which occurred earlier than the expected arrival time of the tsunamis directly generated by the HTHH eruption. In addition to JAGUAR, they used NICAM, a nonhydrostatic AGCM (Sato et al. 2014), with the same background fields and initial perturbations as those used in Watanabe et al. (2022). In their simulations, both models successfully excited Lamb and Pekeris waves that propagated with comparable phase speeds. Furthermore, Suzuki et al. (2023) presented the vertical structures of the meridional wind associated with the Lamb and Pekeris waves along the path from HTHH to Japan approximately five hours after the eruption. The

single node of the Pekeris wave was located near the 75 hPa level in JAGUAR and near the 60 hPa level in NICAM.

In addition to simulations using these AGCMs, many studies have investigated the atmospheric response to the HTHH eruption using simpler atmospheric models. For example, Nishikawa et al. (2022) employed a two-dimensional (horizontal–vertical) atmospheric model on a plane, linearized about a basic state at rest where temperature depends only on altitude, to investigate how atmospheric waves excited by the eruption induced sea level fluctuations. In their experiments, a Gaussian-like heating source with a horizontal standard deviation of 16 km was imposed at an altitude of 40 km. This excited Lamb waves, acoustic waves, and gravity waves, among which the fastest gravity wave can be identified as the Pekeris wave in terms of both phase speed and vertical structure. In contrast, Shinagawa and Miyoshi (2024) used an axisymmetric atmospheric model on a non-rotating sphere to investigate atmospheric variations extending up to the ionosphere. In their experiments, where an extreme temperature increase was imposed at the pole between 0 and 2 km altitudes, Lamb and Pekeris waves were excited and propagated with phase speeds generally consistent with those reported by Watanabe et al. (2022), while at altitudes above 100 km, thermospheric gravity waves with even higher phase speeds became dominant.

As is evident from the substantial differences in the initial perturbations adopted in previous studies, the detailed source structure of atmospheric waves associated with the HTHH eruption remains uncertain. Mizutani and Yomogida (2023) estimated the source altitude required to reproduce the surface pressure fluctuations observed in Japan following the eruption. They showed that the observed fluctuations were better reproduced as a superposition of Lamb and gravity waves when the source altitude was closer to the sea surface. In contrast, Tonegawa and Fukao (2023) showed that Lamb waves and thermospheric gravity waves excited by the HTHH eruption were in resonance, and that, to be consistent with observations, their source altitude must have been between 58 and 70 km. Thus, although these two studies are both based on the method of Harkrider (1964), their estimated source structures for the atmospheric waves associated with the HTHH eruption differ significantly.

Having outlined the observational and numerical studies of the Pekeris wave, we now turn to a brief review of studies addressing its theoretical aspects. In an isothermal stationary atmosphere, the Lamb wave has the same equivalent depth and vertical structure as the Lamb mode, which is the only homogeneous solution of the vertical structure equation (VSE) in classical atmospheric tidal theory that satisfies the rigid boundary condition at the bottom and the evanescent condition at the top. Pekeris (1937), at a time when the vertical temperature structure of the atmosphere was not yet

well understood, solved the VSE assuming the presence of a high-temperature region near the stratopause, and demonstrated that, in addition to the Lamb mode, another vertical mode could exist, which is now known as the Pekeris mode. Subsequently, Salby (1979) solved the VSE using a more realistic vertical temperature profile based on the US Standard Atmosphere (NOAA, NASA, USAF 1976; hereafter referred to as USSA76), and showed the vertical structures and equivalent depths of these two vertical modes for the real atmosphere. It should be noted that the equivalent depth corresponding to the Pekeris mode differs significantly from the counterpart obtained by Pekeris (1937), due to the discrepancy in the assumed background vertical temperature profile. Furthermore, although Pekeris (1937) and Salby (1979) considered normal modes on a rotating sphere and solved the VSE to determine the equivalent depths for them, the VSE itself is applicable to any free modal waves regardless of the system rotation and horizontal scale. Consequently, among localized acoustic-gravity waves, those denoted specifically as the  $L_0$  and  $L_1$  modes in Francis (1973) substantially correspond to the Lamb and Pekeris modes derived via the VSE.

Following the demonstration by Watanabe et al. (2022) of the existence of atmospheric waves exhibiting the characteristics of the Pekeris mode shown by Salby (1979), Ishioka (2023) revisited solutions of the VSE by addressing issues in the calculation method of Salby (1979). The equivalent depths calculated using the vertical temperature profile of USSA76 were 9.9 km for the Lamb mode and 6.6 km for the Pekeris mode. These values differ from those estimated by Watanabe et al. (2022), particularly for the Pekeris mode. In addition, the node of the log-pressure velocity (log- $p$  velocity) of the Pekeris mode was located at 10.3 km (approximately 253 hPa in USSA76), which differs substantially from the simulation result of Watanabe et al. (2022).

One possible cause of the differences in the equivalent depth and the node altitude of the Pekeris mode between Watanabe et al. (2022) and Ishioka (2023) is the difference between the vertical temperature structure of USSA76 and that in the vicinity of HTHH at the time of the eruption. Therefore, Ishizaki et al. (2023) solved the VSE using vertical temperature profiles obtained by averaging ERA5 reanalysis data (Hersbach et al. 2020) over various periods and regions, and compared the results. The equivalent depths calculated using the vertical temperature profile in the vicinity of HTHH at the time of the eruption were approximately 10.1 km for the Lamb mode and 6.5 km for the Pekeris mode. Compared with Watanabe et al. (2022), the equivalent depth of the Lamb mode shows good agreement, while a discrepancy remains for the Pekeris mode. In addition, the node of the log- $p$  velocity of the Pekeris mode

obtained using the globally averaged temperature profile for January 2022 was located at approximately 270 hPa, which differs markedly from the result of Watanabe et al. (2022).

The primary objective of the present study is to clarify the causes of the discrepancies in the equivalent depth and the node altitude of the Pekeris mode reported by Watanabe et al. (2022) and Ishizaki et al. (2023). One possible reason for these discrepancies is that a few hours after the HTHH eruption, the Lamb wave existed as an isolated wave owing to its fast phase speed, while the Pekeris wave may have been mixed with other disturbances. However, as described above, the detailed source structure of atmospheric waves during the HTHH eruption is not well known. Therefore, in the present study, we propose a relatively simple atmospheric modeling framework that allows atmospheric waves excited by different structures of initial temperature perturbations to be calculated and compared. First, for ease of comparison with Watanabe et al. (2022), we employ the hydrostatic approximation. In addition, to reduce computational costs, we assume axisymmetry on a non-rotating sphere following Shinagawa and Miyoshi (2024), and we perform linear calculations about a basic state at rest where temperature depends only on altitude. Using this model, we investigate the characteristics of Lamb and Pekeris waves excited by various initial perturbations and compare them with observations.

The equivalent depth and vertical structure of the Pekeris mode may also be substantially modified by the effects of background winds. Therefore, we also perform linear calculations using a two-dimensional (horizontal-vertical) model on a plane, as in Nishikawa et al. (2022). Unlike their study, however, we adopt the hydrostatic approximation and assume a basic state in which both temperature and zonal wind depend on altitude. Using this model, we investigate the effects of background winds on Lamb and Pekeris waves. Furthermore, following the method of Ishioka (2026), we calculate Lamb and Pekeris mode solutions modified by the background winds which satisfy either the evanescent condition or the radiation condition, and examine their underlying physical mechanisms.

The remainder of this paper is organized as follows. In Sect. 2, we describe the details of the reanalysis and satellite observation data used in the present study. The numerical methods for calculating atmospheric waves excited by a temperature perturbation, as well as the method for computing the Lamb and Pekeris mode solutions in the presence of background winds, are described in Sect. 3. In the same section, we also describe the method used to convert temperature disturbances into the second derivative of radiance for comparison with observations. The results are presented in Sect. 4. A summary and discussion are given in Sect. 5.

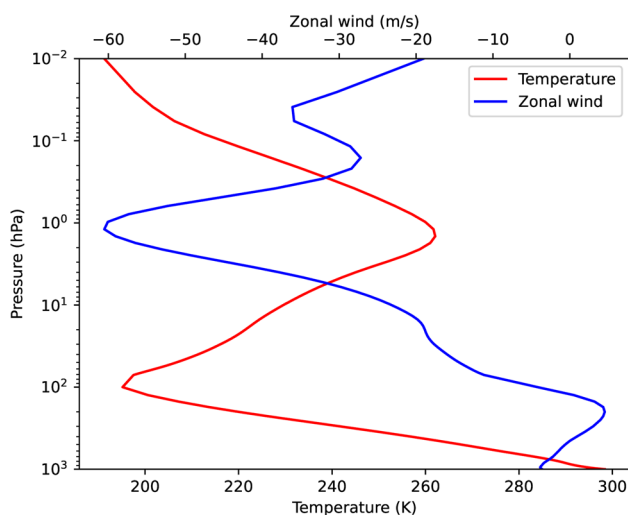
## 2 Data

In this study, we utilize ERA5 reanalysis data and satellite observations from Himawari-8 and GOES-17, as detailed below.

### 2.1 ERA5

As the background state for our numerical simulations, we use temperature and zonal wind data from ERA5, the latest atmospheric reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). Since the Pekeris mode is highly sensitive to the vertical temperature structure of the mesosphere (Ishizaki et al. 2023), the ERA5 pressure-level data (Hersbach et al. 2023), which extend only up to 1 hPa, are insufficient. Therefore, we supplement them with model-level data (Hersbach et al. 2017). Following the procedure in Ishizaki et al. (2023), model-level data on and above 71.1187 hPa and pressure-level data on and below 100 hPa are merged to construct an 81-level dataset ranging from 1000 hPa to 0.01 hPa. The horizontal grid interval is  $1^\circ \times 1^\circ$  for both datasets. We use monthly averaged data for January 2022, the month of the HTHH eruption.

In our numerical experiments, the background field is assumed to vary only with altitude. We use vertical profiles of temperature and zonal wind obtained by zonally averaging the ERA5 dataset at  $20^\circ\text{S}$  (Fig. 1), considering that HTHH is situated at  $20.55^\circ\text{S}$  and previous simulations (Watanabe et al. 2022) showed vertical profiles of Lamb and Pekeris waves averaged over  $20^\circ\text{--}21^\circ\text{S}$ . The profiles extend to 0.01 hPa, above which the atmosphere is assumed to be isothermal and the zonal wind uniform. At 0.01 hPa, the



**Fig. 1** Vertical profiles of temperature and zonal wind, obtained by zonally averaging ERA5 monthly mean data for January 2022 at  $20^\circ\text{S}$ . The color legend is shown in the figure

temperature is approximately 190 K; using this profile in the VSE allows for the application of the evanescent condition at the top boundary for both Lamb and Pekeris modes (Ishizaki et al. 2023). While considering the thermospheric temperature increase would make both modes wavy above a certain altitude, these effects of upward energy leakage have been addressed by Ishioka (2026) and are neglected here. The background zonal wind (Fig. 1) shows a weak westerly region below 100 hPa, but easterlies dominate at most altitudes, particularly around 1 hPa.

### 2.2 Himawari-8

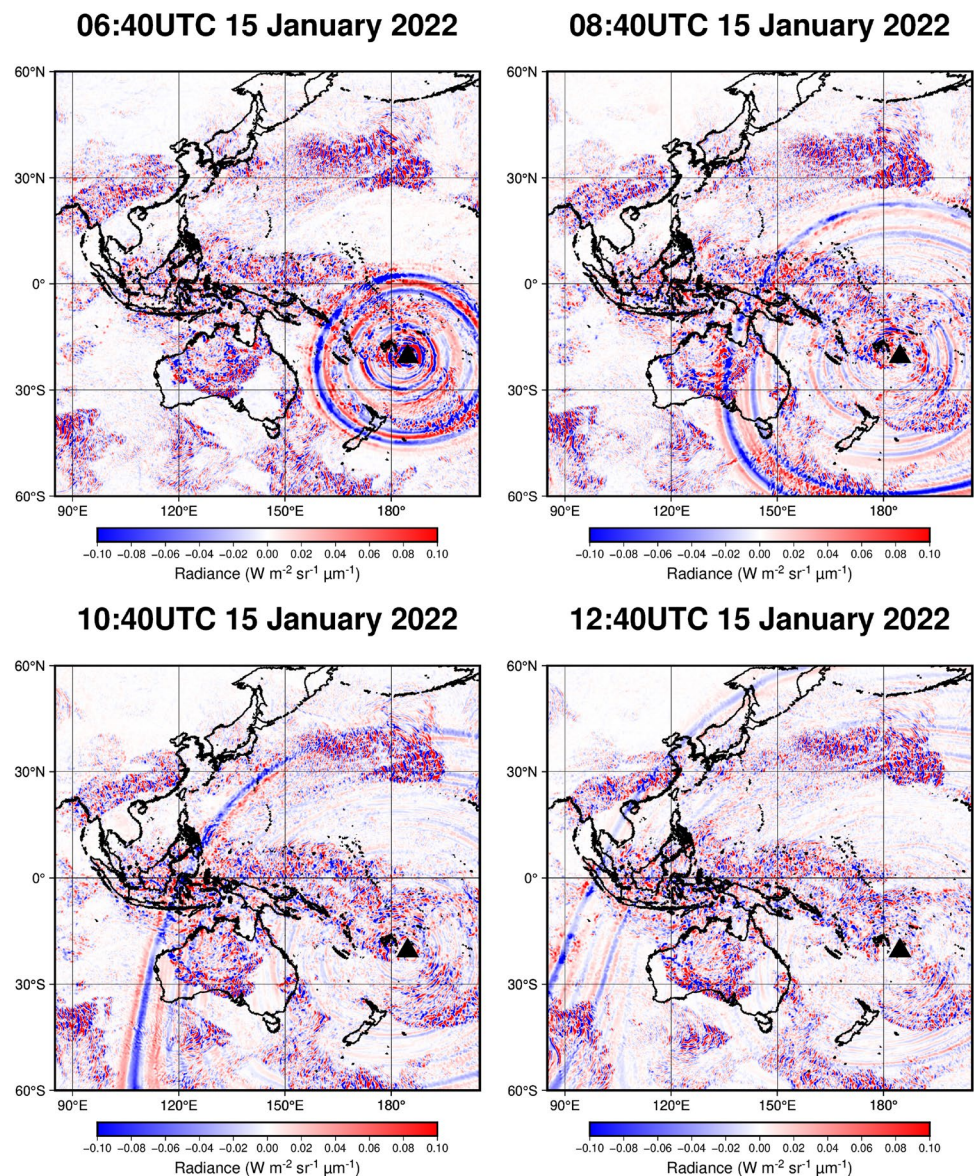
We visualize the Lamb and Pekeris waves from satellite observations to compare them with numerical results. Satellite data are essential because the Pekeris wave's amplitude in the troposphere is significantly smaller than that of the Lamb wave, making its detection via ground-based barometers challenging (Watanabe et al. 2022; Rose et al. 2024). Based on the finding by Murakami et al. (2026) that Band 12 is optimal for detecting Pekeris waves, we use radiance data from Band 12 of Himawari-8, provided by the Center for Environmental Remote Sensing (CERES), Chiba University (Takenaka et al. 2020; Yamamoto et al. 2020). The data cover  $85^\circ\text{E}\text{--}155^\circ\text{W}$  and  $60^\circ\text{N}\text{--}60^\circ\text{S}$  with a  $0.04^\circ \times 0.04^\circ$  grid and 10-minute temporal resolution.

To isolate the waves, we calculate the second-order time difference of radiance,  $I''_*$ . Here and throughout this paper, variables accompanied by the subscript “\*” (such as  $I_*$ ) represent dimensional quantities, to distinguish them from the dimensionless variables introduced in later sections. The second-order difference is defined as

$$I''_* = I_*^+ - 2I_*^0 + I_*^-, \quad (1)$$

where  $I_*$  is the radiance, and the superscripts 0, +, and – denote the values at a given reference observation time and at the times immediately following and preceding it, respectively. The interval between these successive observations is 10 min. As demonstrated by Otsuka (2022), this second-order difference clarifies wavefronts. Furthermore, this procedure suppresses contamination from linear surface temperature variations. Figure 2 shows  $I''_*$  distributions at approximately 2, 4, 6, and 8 h after the eruption, with a Gaussian filter applied to suppress convective noise. While the wave type is difficult to identify at 2 h, the Lamb and Pekeris waves are clearly distinguishable with comparable amplitudes by 4 h, once internal gravity waves have decayed. While the wave type is difficult to identify at 2 h, two distinct wavefronts with comparable amplitudes emerge by 4 h, once internal gravity waves have decayed; the leading wavefront corresponds to the Lamb wave, followed by

**Fig. 2** Distributions of second-order time difference of radiance observed in Band 12 of Himawari-8 at approximately 2, 4, 6, and 8 h after the HTHH eruption. The time interval is 10 min, and a Gaussian filter is applied to clarify the wavefronts of Lamb and Pekeris waves. The triangular symbol indicates the location of the HTHH



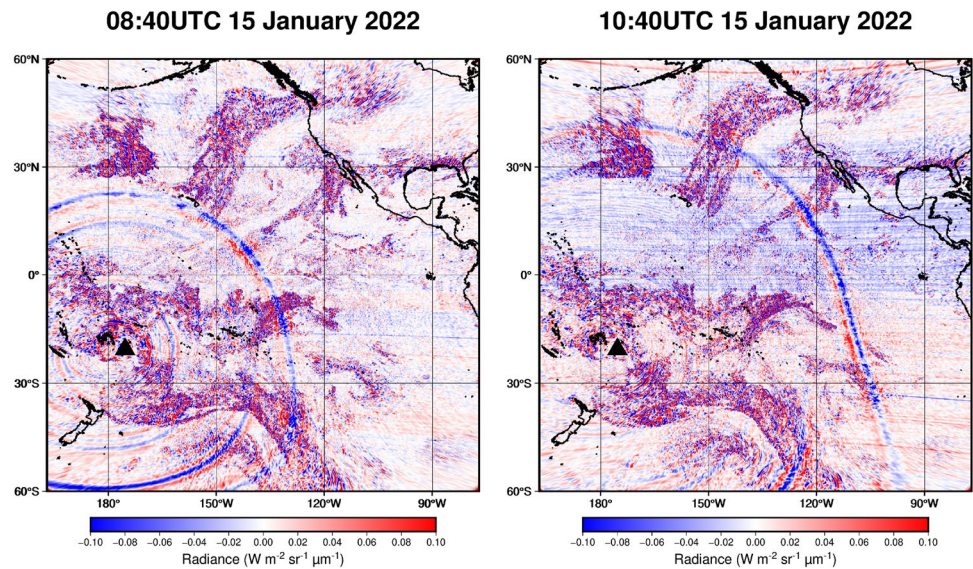
the Pekeris wave as the second wavefront. These wavefronts are notably sharper than those obtained using first-order differences (cf. Figure 1 of Watanabe et al. 2022). By 6–8 h, the Pekeris wave amplitude slightly decreases relative to the Lamb wave, and the northward-propagating Pekeris wave appears weaker than the westward-propagating one.

### 2.3 GOES-17

Since the eastward-propagating waves fall outside Himawari-8's field of view, we use data from the GOES-17 geostationary satellite (located at 137°W). This data is provided by National Oceanic and Atmospheric Administration (NOAA). We analyze Band 12 radiance, which shares the same central wavelength as Himawari-8's Band 12. The spatial resolution is 0.000056 radians (nadir viewing

angle), and the temporal resolution is 10 min. Due to poor data quality at 8 h post-eruption, Fig. 3 shows  $I''_*$  only at 4 and 6 h. Despite some instrumental striping and convective noise, both wave types are visible. Comparing Figs. 2 and 3 reveals that the eastward-propagating Pekeris wave is significantly weaker than those propagating in other directions, a result consistent with the simulation findings of Watanabe et al. (2022).

**Fig. 3** Same as Fig. 2, except that the radiance is observed in Band 12 of GOES-17 and only those at approximately 4 and 6 h after the eruption are shown



### 3 Numerical calculation methods

#### 3.1 Formulation for a spherical atmosphere without background winds

In the present study, the atmospheric response to an initial perturbation representing a volcanic eruption is examined using a linear atmospheric model. As the governing equations, we employ the primitive equations in  $\log-p$  coordinates on a non-rotating sphere (e.g., Andrews et al., 1987), linearized around a motionless basic state where the background temperature depends only on altitude. Let  $p_*$  be the pressure and  $p_{0*}$  be the reference surface pressure; the vertical coordinate is defined as  $z = -\ln(p_*/p_{0*})$ . To eliminate the effect of the upward decrease in atmospheric density, the geopotential and  $\log-p$  velocity perturbations are scaled by  $e^{z/2}$ , and the resulting scaled variables are denoted by  $\phi$  and  $W$ , respectively. Subsequently, the equations are nondimensionalized using the reference temperature  $T_*$ , the scale height  $H_*$  ( $= R_* T_*/g_*$ , where  $R_*$  is the gas constant and  $g_*$  is the gravitational acceleration), and the time scale  $H_*/\sqrt{R_* T_*}$ . The resulting nondimensionalized governing equations are expressed as follows:

$$\frac{\partial}{\partial t} \left( \frac{\partial}{\partial z} + \frac{1}{2} \right) \phi = -N^2 W, \quad (2)$$

$$\frac{\partial}{\partial t} \left( \frac{\partial}{\partial z} - \frac{1}{2} \right) W = \nabla^2 \phi. \quad (3)$$

Here,  $t$  is the time, and the squared  $\log-p$  buoyancy frequency  $N^2$  is written as

$$N^2 = \frac{d\bar{T}}{dz} + \kappa \bar{T}, \quad (4)$$

where  $\kappa = R_*/C_{p*}$  (with  $C_{p*}$  being the specific heat at constant pressure), and  $\bar{T}$  is the dimensionless background temperature field, which depends only on the  $\log-p$  coordinate  $z$ . The nondimensional differential operator  $\nabla^2$  is defined as

$$\nabla^2 = \frac{H_*^2}{a_*^2} \left\{ \frac{1}{1-\mu^2} \frac{\partial^2}{\partial \lambda^2} + \frac{\partial}{\partial \mu} \left[ (1-\mu^2) \frac{\partial}{\partial \mu} \right] \right\}, \quad (5)$$

where  $a_*$  is the radius of the Earth,  $\lambda$  is the longitude, and  $\mu = \cos \theta$ , with  $\theta$  being the colatitude.

As the lower boundary condition, we impose the requirement that the vertical velocity vanishes at the surface:

$$\frac{\partial}{\partial t} \phi + \bar{T} W = 0 \quad (z = 0). \quad (6)$$

Since it is generally impossible to impose a radiation or evanescent upper boundary condition without employing Fourier transforms, we place the upper boundary at  $z = z_{ub}$  and impose the condition that the  $\log-p$  velocity perturbation vanishes there:

$$W = 0 \quad (z = z_{ub}). \quad (7)$$

In the present study, we analyze the atmospheric response to an axisymmetric initial perturbation imposed at the North Pole. Under this formulation, the assumption ensures the longitudinal uniformity of the generated disturbances, and each variable is expanded in terms of Legendre polynomials as follows:

$$\phi(z, \mu, t) = \sum_{n=0}^M \hat{\phi}_n(z, t) P_n(\mu), \quad (8)$$

$$W(z, \mu, t) = \sum_{n=0}^M \hat{W}_n(z, t) P_n(\mu). \quad (9)$$

Here, the hat symbol (^) denotes the expansion coefficients,  $M$  is the truncation wavenumber, and  $P_n(\mu)$  is defined as

$$P_n(\mu) = \sqrt{2n+1} \frac{1}{2^n n!} \frac{d^n}{d\mu^n} (\mu^2 - 1)^n \quad (n \geq 0). \quad (10)$$

By substituting these expansions, (2) and (3), together with the boundary conditions (6) and (7), can be expressed separately for each total wavenumber  $n$  as follows:

$$\frac{\partial}{\partial t} \left( \frac{\partial}{\partial z} + \frac{1}{2} \right) \hat{\phi}_n = -N^2 \hat{W}_n, \quad (11)$$

$$\frac{\partial}{\partial t} \left( \frac{\partial}{\partial z} - \frac{1}{2} \right) \hat{W}_n = \frac{-n(n+1)H_*^2}{a_*^2} \hat{\phi}_n, \quad (12)$$

$$\frac{\partial}{\partial t} \hat{\phi}_n + \bar{T} \hat{W}_n = 0 \quad (z = 0), \quad (13)$$

$$\hat{W}_n = 0 \quad (z = z_{ub}). \quad (14)$$

These equations are discretized in the vertical direction using the finite-difference method described by Ishioka (2026). The time integration of (11)–(13) is performed using a fourth-order Runge–Kutta scheme.

As an initial condition, a temperature perturbation  $T'$  with a Gaussian profile in both the vertical and horizontal directions is imposed at the log- $p$  coordinate  $z = z_0$  and colatitude  $\theta = 0$ , as follows:

$$T'(z, \theta) = \Delta T \exp \left( \frac{-(z - z_0)^2}{2\sigma_z^2} \right) e^{\alpha(\cos \theta - 1)}. \quad (15)$$

Here,  $\Delta T$  is the amplitude of the initial temperature perturbation at its center,  $\sigma_z$  is the standard deviation of the Gaussian profile in the vertical direction, and  $\alpha$  is a parameter that controls the horizontal extent of the perturbation. If one strictly aims to reproduce the actual eruption process, one should specify the size of the heat source and vary  $\Delta T$  with altitude in accordance with the background atmospheric density. In the present study, however,  $\Delta T$  is set as a constant to facilitate a straightforward comparison with Watanabe et al. (2022), which imposed a temperature perturbation. Note that if one wishes to quantitatively evaluate the effect of varying the magnitude of  $\Delta T$ , the resulting

responses need only be scaled with altitude as appropriate, owing to the linear nature of the numerical experiments. To compute the time evolution of (11)–(13), initial values of  $\phi$  and  $W$  are required. The temperature perturbation  $T'$  is converted into the unscaled geopotential perturbation  $\phi'$  ( $= \phi e^{z/2}$ ) using the following hydrostatic relation:

$$\frac{\partial \phi'}{\partial z} = T'. \quad (16)$$

Assuming the initial temperature perturbation has the Gaussian profile defined above, the unscaled geopotential perturbation at  $t = 0$  is given by

$$\begin{aligned} \phi'(t = 0, z, \mu) &= \phi'(t = 0, z = 0, \mu) \\ &+ \Delta T e^{\alpha(\mu-1)} \int_0^z \exp \left( \frac{-(z' - z_0)^2}{2\sigma_z^2} \right) dz' \\ &= \phi'(t = 0, z = 0, \mu) \\ &+ \Delta T e^{\alpha(\mu-1)} \frac{\sqrt{\pi}\sigma_z}{\sqrt{2}} \left[ \operatorname{erf} \left( \frac{z - z_0}{\sqrt{2}\sigma_z} \right) + \operatorname{erf} \left( \frac{z_0}{\sqrt{2}\sigma_z} \right) \right], \end{aligned} \quad (17)$$

where the error function is defined as  $\operatorname{erf}(z) = \frac{2}{\sqrt{\pi}} \int_0^z e^{-z'^2} dz'$ . There remains arbitrariness in specifying the surface value  $\phi'(t = 0, z = 0, \mu)$ , which we set to zero in the present study. The initial log- $p$  velocity perturbation is also set to zero.

The parameter settings used in the calculation are as follows:  $R_* = 287 \text{ m}^2 \text{ s}^{-2} \text{ K}^{-1}$ ,  $p_{0*} = 1000 \text{ hPa}$ ,  $\kappa = 2/7$ ,  $a_* = 6.37 \times 10^6 \text{ m}$ ,  $g_* = 9.81 \text{ m s}^{-2}$ ,  $T_* = 273.15 \text{ K}$ , and  $\Delta T \times T_* = 100 \text{ K}$  (noting that the value of  $T_*$  does not affect the results). The number of latitudinal grid points is 512,  $M$  is 511,  $z_{ub} = 200$ , the vertical grid spacing is 0.1. The dimensionless time step corresponds to a physical time of 30 s, and the integration is carried out up to an equivalent of 8 h. By setting the model top extremely high in this way, the influence of reflections from the model top does not appear below the mesopause during the analysis period, so no artificial sponge layer or dissipation is applied. In the present study, we fix  $\sigma_z = 1$ , and only the results with varying  $\alpha$  and  $z_0$  are presented.

### 3.2 Formulation for a planar atmosphere with background winds

The method described in Sect. 3.1 assumes a motionless atmosphere as the basic state; in this subsection, we extend the formulation to include the effects of background winds. For simplicity, we neglect the effects of spherical geometry and consider two-dimensional atmospheric motion in the zonal and vertical directions with a prescribed zonal wind  $\bar{U}$ , which depends only on the log- $p$  coordinate. Note that

neglecting the earth's curvature results in omitting the metric acceleration terms in the governing equations on spherical geometry. However, since the present study focuses on short-term events of about several hours, these terms can be reasonably omitted, just as the Earth's rotation has already been omitted. Under these assumptions, (2) and (3), as well as the lower boundary condition (6), are modified as follows:

$$\left(\frac{\partial}{\partial t} + \bar{U} \frac{\partial}{\partial x}\right) \left(\frac{\partial}{\partial z} + \frac{1}{2}\right) \phi = -N^2 W, \quad (18)$$

$$\left(\frac{\partial}{\partial t} + \bar{U} \frac{\partial}{\partial x}\right) \left(\frac{\partial}{\partial z} - \frac{1}{2}\right) W - \frac{\partial \bar{U}}{\partial z} \frac{\partial W}{\partial x} = \frac{\partial^2 \phi}{\partial x^2}, \quad (19)$$

$$\left(\frac{\partial}{\partial t} + \bar{U} \frac{\partial}{\partial x}\right) \phi + \bar{T} W = 0 \quad (z = 0). \quad (20)$$

Here,  $x$  denotes the nondimensionalized coordinate in the zonal direction. The upper boundary condition remains the same as that used in Sect. 3.1.

We impose periodic boundary conditions in the zonal direction and expand each variable as a Fourier series:

$$\phi(z, x, t) = \sum_{k=-K}^K \operatorname{Re} \left[ \hat{\phi}_k(z, t) \exp \left( \frac{2\pi i k x}{x_L} \right) \right], \quad (21)$$

$$W(z, x, t) = \sum_{k=-K}^K \operatorname{Re} \left[ \hat{W}_k(z, t) \exp \left( \frac{2\pi i k x}{x_L} \right) \right], \quad (22)$$

where  $K$  is the truncation wavenumber,  $x_L$  is the total horizontal width of the domain, and the hat symbol again denotes the Fourier expansion coefficients. Then, (18) and (19), together with the boundary condition (20), can be expressed separately for each zonal wavenumber  $k$  as follows:

$$\left(\frac{\partial}{\partial t} + i\bar{k}\bar{U}\right) \left(\frac{\partial}{\partial z} + \frac{1}{2}\right) \hat{\phi}_k = -N^2 \hat{W}_k, \quad (23)$$

$$\left(\frac{\partial}{\partial t} + i\bar{k}\bar{U}\right) \left(\frac{\partial}{\partial z} - \frac{1}{2}\right) \hat{W}_k - i\bar{k} \frac{\partial \bar{U}}{\partial z} \hat{W}_k = -\bar{k}^2 \hat{\phi}_k, \quad (24)$$

$$\left(\frac{\partial}{\partial t} + i\bar{k}\bar{U}\right) \hat{\phi}_k + \bar{T} \hat{W}_k = 0 \quad (z = 0), \quad (25)$$

where  $\bar{k} = 2\pi k/x_L$ . The vertical discretization is performed using the same finite-difference approximation as described in Sect. 3.1. Time integration is also carried out following the identical procedure.

As in Sect. 3.1, the initial perturbation is given as a Gaussian temperature anomaly in both the zonal and vertical directions, centered at  $z = z_0$  and  $x = 0$ , as follows:

$$T'(z, x) = \Delta T \exp \left( \frac{-(z - z_0)^2}{2\sigma_z^2} \right) \exp \left( \frac{-x^2}{2\sigma_x^2} \right), \quad (26)$$

where  $\sigma_x$  is the standard deviation in the zonal direction. In these experiments,  $\sigma_x \times H_* = 140$  km, the number of zonal grid points is 1024,  $K$  is 511, and the horizontal domain  $x_*$  ( $= x \times H_*$ ) ranges from  $-20000$  km to  $20000$  km (corresponding to  $x_L \times H_* = 40000$  km). All other parameters are the same as those used in Sect. 3.1.

### 3.3 Method for calculating Lamb and Pekeris mode solutions under the influence of background winds

In this section, we describe the numerical method for determining the phase speed  $c$  of Lamb and Pekeris modes, following the approach of Ishioka (2026). Assuming the zonal and time dependence of each perturbation as

$$\phi(x, z, t) = \operatorname{Re}[\hat{\phi}(z)e^{ik(x-ct)}], \quad (27)$$

$$W(x, z, t) = \operatorname{Re}[\hat{W}(z)e^{ik(x-ct)}], \quad (28)$$

the governing equations (18) and (19) are rewritten as a system of first-order ordinary differential equations for the vertical structures:

$$\left(\frac{\partial}{\partial z} + \frac{1}{2} - \frac{1}{\bar{c}} \frac{\partial \bar{U}}{\partial z}\right) \hat{\Phi} = -\frac{N^2 \hat{W}}{\bar{c}^2}, \quad (29)$$

$$\left(\frac{\partial}{\partial z} - \frac{1}{2} + \frac{1}{\bar{c}} \frac{\partial \bar{U}}{\partial z}\right) \hat{W} = \hat{\Phi}, \quad (30)$$

where  $\bar{c} = c - \bar{U}$  and  $\hat{\Phi} = -ik\hat{\phi}/\bar{c}$ . These equations are solved subject to the bottom boundary condition at  $z = 0$ :

$$\hat{\Phi} + \frac{\bar{T} \hat{W}}{\bar{c}^2} = 0 \quad (z = 0). \quad (31)$$

To derive an appropriate upper boundary condition, we assume that the background temperature  $\bar{T}$  and wind  $\bar{U}$  are constant above  $z_{\text{top}}$ . Here,  $z_{\text{top}}$  is the log- $p$  coordinate corresponding to 0.01 hPa, which is the uppermost level of the dataset used in the present study. In this region,  $N^2 = \kappa \bar{T}$  and  $\partial \bar{U} / \partial z = 0$ . By substituting (30) into (29), the system can be reduced to a single second-order differential equation for  $\hat{W}$ :

$$\frac{d^2 \hat{W}}{dz^2} + \left( \frac{\kappa \bar{T}}{\bar{c}^2} - \frac{1}{4} \right) \hat{W} = 0. \quad (32)$$

The general solution of (32) is proportional to  $e^{sz}$ , where  $s^2 = 1/4 - \kappa\bar{T}/\bar{c}^2$ . The choice of the branch for  $s$  depends on the physical characteristics of the mode and the range of  $c$ .

First, we search for  $c$  within the range of real numbers to find “trapped” modes. For a solution to be a trapped mode, it must satisfy the evanescent condition at the upper boundary, requiring the wave amplitude to decay as  $z \rightarrow \infty$ . Thus, the vertical structure at  $z \geq z_{\text{top}}$  is given by

$$\hat{W} \propto \exp\left[-z\sqrt{\frac{1}{4} - \frac{\kappa\bar{T}}{\bar{c}^2}}\right] \quad \text{for} \quad \frac{1}{4} - \frac{\kappa\bar{T}}{\bar{c}^2} > 0. \quad (33)$$

If no real solution for  $c$  is found, we extend the search to the complex plane to account for modes that may decay in time due to upward energy leakage. In this case, we must impose the radiation boundary condition to allow for the upward propagation of energy. The corresponding vertical structure is written as

$$\hat{W} \propto \exp\left[-iz\sqrt{\frac{\kappa\bar{T}}{\bar{c}^2} - \frac{1}{4}\text{sgn}(\text{Re}[\bar{c}])}\right] \quad \text{for} \quad \text{Re}\left[\frac{\kappa\bar{T}}{\bar{c}^2} - \frac{1}{4}\right] > 0, \quad (34)$$

where the branch of the square root is chosen such that its real part is positive to represent outgoing waves.

The numerical procedure is as follows. We integrate (29) and (30) downward from  $z_{\text{top}}$  using either (33) or (34) as the initial condition. The phase speed  $c$  is determined via the Newton method so that the bottom boundary condition (31) is satisfied. Note that the equivalent depth can be readily determined as needed using the dispersion relation for shallow water waves. All other parameters are consistent with those described in Sect. 3.1.

### 3.4 Method for converting temperature disturbances into the second derivative of radiance

In this section, we describe a method for converting temperature disturbances into the second derivative of radiance, which facilitates direct comparison between satellite observations and numerical calculations. In the following discussion, the surface and air temperatures  $T_s$  and  $T_a$ , respectively, as well as the background temperature  $\bar{T}$  are treated as dimensionless variables normalized by the reference temperature  $T_*$ . For a specific wavelength  $\lambda_*$ , the Planck function  $B_*(T)$  is defined as a function of the dimensionless temperature  $T$  as follows:

$$B_*(T) = \frac{2hP_*c_*^2}{\lambda_*^5} \frac{1}{\exp\left(\frac{hP_*c_*}{\lambda_*k_{B_*}TT_*}\right) - 1}, \quad (35)$$

where  $h_{P_*} = 6.626 \times 10^{-34}$  J s is the Planck constant,  $c_* = 2.998 \times 10^8$  m s<sup>-1</sup> is the speed of light, and  $k_{B_*} = 1.381 \times 10^{-23}$  J K<sup>-1</sup> is the Boltzmann constant. This formula represents the monochromatic radiance emitted at a single wavelength  $\lambda_*$ , representing the central wavelength of an observation band.

The radiance  $I_*$  observed at the top of the atmosphere is expressed as the sum of contributions from the radiation emitted by the Earth’s surface and the atmosphere:

$$I_* = B_*(T_s)\tau + \int_0^\infty B_*(T_a)F(z)dz. \quad (36)$$

Here,  $\tau$  is the transmittance from the surface to the top of the atmosphere, and  $F(z)$  is the emission weighting function. By definition, they satisfy the following relationship:

$$1 - \tau = \int_0^\infty F(z)dz. \quad (37)$$

As described in Sect. 2.2, the second derivative of the radiance emitted from the Earth’s surface is small, so the first term in (36) is neglected. Under this assumption, by expanding  $B_*(T_a)$  in a Taylor series around the dimensionless background temperature  $\bar{T}$ :

$$I_* \approx \int_0^\infty \left[ B_*(\bar{T}) + \frac{\partial B_*}{\partial T} \Big|_{\bar{T}} T' \right] F(z)dz, \quad (38)$$

and taking the second-order temporal difference as (1), we obtain

$$I_*'' \approx \int_0^\infty \frac{\partial B_*}{\partial T} \Big|_{\bar{T}} (T'^+ - 2T'^0 + T'^-) F(z)dz. \quad (39)$$

For comparison with Himawari-8 Band 12, we set the central wavelength  $\lambda_* = 9.6 \mu\text{m}$ . In this study, we approximate the weighting function  $F(z)$  based on the clear-sky radiative transfer calculation for Band 12 by Zou et al. (2016):

$$F(z) = 0.4e^{-3z} + 0.2 \exp\left[-\frac{(z + \ln(p_{c_*}/p_{0_*}))^2}{2}\right], \quad (40)$$

where  $p_{c_*} = 40$  hPa. This vertical profile yields a total transmittance of  $\tau \approx 0.34$ , which is consistent with the transmittance for a clear-sky atmosphere without aerosols e.g., (Petty, 2006).

## 4 Results

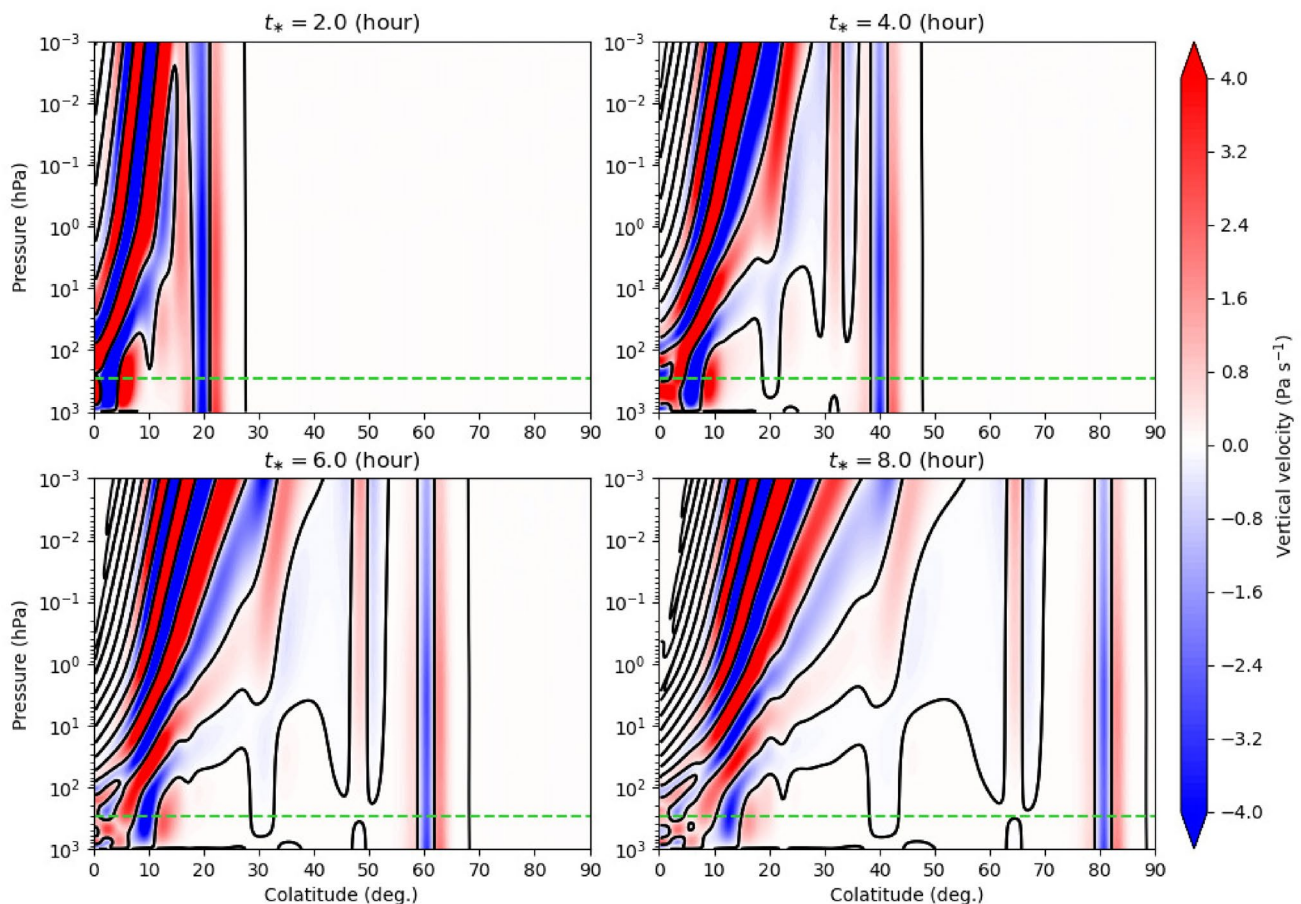
### 4.1 Results on a sphere without background winds

First, we present results obtained using the method described in Sect. 3.1, with  $\alpha = 2 \times 10^3$  and  $z_0 = 1$ . The value  $\alpha = 2 \times 10^3$  corresponds to a horizontal standard deviation of approximately 142 km for the initial Gaussian-like perturbation, while  $z_0 = 1$  represents its central altitude at the 368 hPa level. These parameters are chosen to roughly match those adopted in Watanabe et al. (2022). Figure 4 shows colatitude–height cross sections of the dimensional  $p$ -velocity  $\omega_*$ , normalized by  $(p_{0*}/p_*)^{5/7}$ , at dimensional times  $t_* = 2, 4, 6$ , and 8 h. This normalization follows Watanabe et al. (2022) so that the amplitude of a Lamb wave remains constant with height in an isothermal atmosphere. The dimensional  $p$ -velocity is given by

$$\omega_* = -\frac{\sqrt{R_* T_*}}{H_*} p_* W e^{z/2}, \quad (41)$$

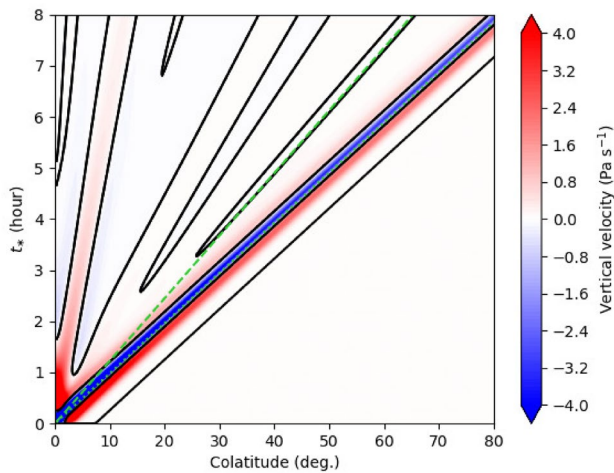
where  $W$  is the scaled log- $p$  velocity perturbation. At  $t_* = 2$  h, a wave with a vertically uniform phase — a Lamb wave — is observed at  $\theta \approx 20^\circ$ . At  $t_* = 4$  h, following the Lamb wave, another wave appears at  $\theta \approx 32^\circ$  whose phase reverses once in the vertical direction. As time progresses, the altitude of this phase reversal approaches the theoretical node altitude for the Pekeris mode, confirming that the wave corresponds to the Pekeris wave. Under these settings, the amplitude of the Pekeris wave is smaller than that of the Lamb wave. Note that because this model is hydrostatic, acoustic waves are not generated; instead, the signals following the Lamb and Pekeris waves are internal gravity waves propagating both horizontally and vertically.

The fact that the wave following the Lamb wave is a Pekeris wave is also confirmed by its phase speed. Figure 5 shows a colatitude–time cross section of  $\omega_*$  at the surface ( $z = 0$ ). The Lamb wave propagates fastest, and its phase speed is in good agreement with the theoretical value. Furthermore, the second wave closely coincides with the theoretical phase speed of the Pekeris wave.



**Fig. 4** Colatitude–height cross sections of the normalized  $p$ -velocity  $\omega_* \times (p_{0*}/p_*)^{5/7}$  at  $t_* = 2, 4, 6$ , and 8 h, for the case of  $\alpha = 2 \times 10^3$  and  $z_0 = 1$ , using the method described in Sect. 3.1. The black contour

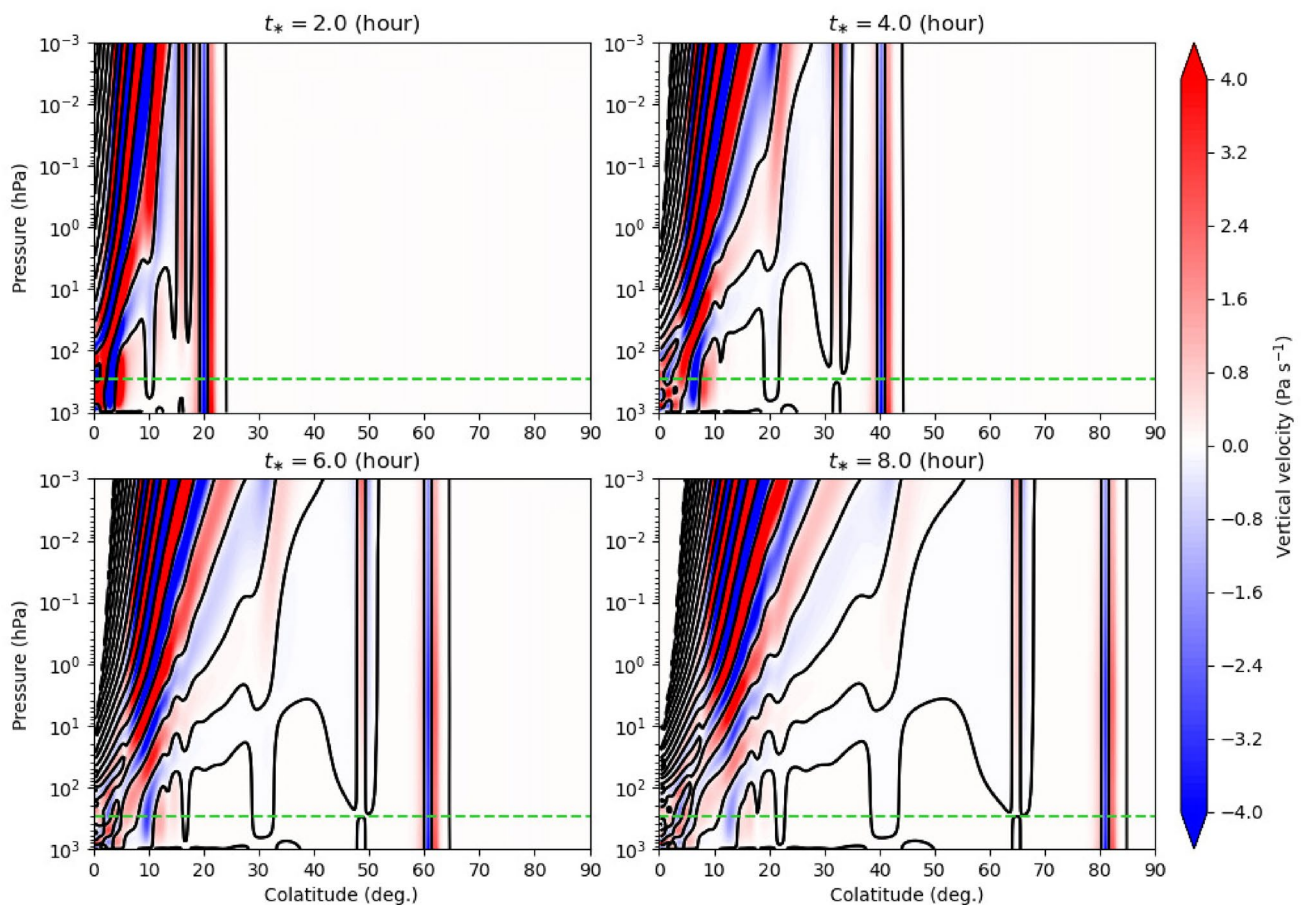
lines are drawn at  $\pm 0.00001 \text{ Pa s}^{-1}$ . The green dashed line indicates the node altitude of the Pekeris wave obtained by solving the VSE using the vertical temperature profile shown in Fig. 1



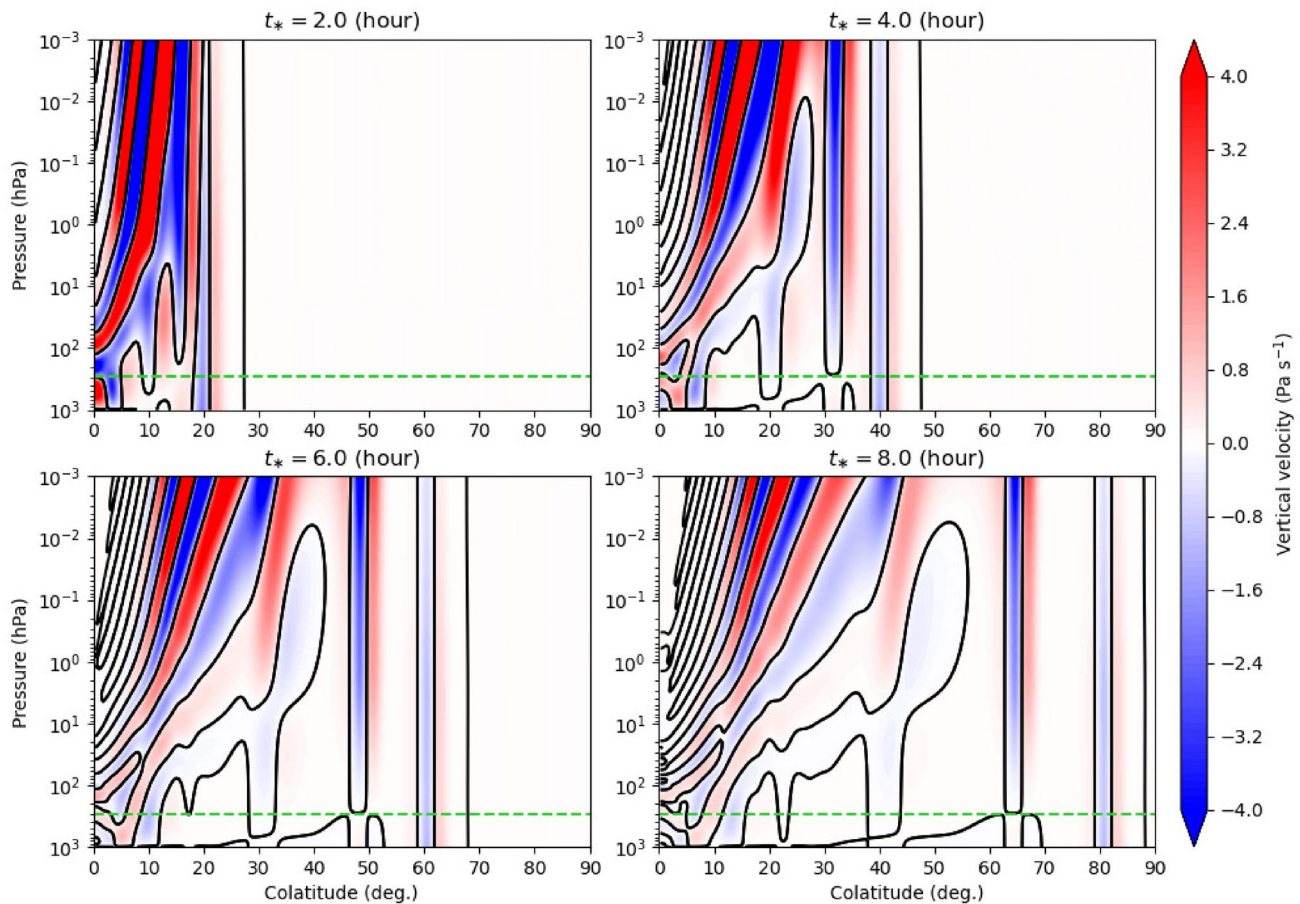
**Fig. 5** The colatitude–time cross section of the  $p$ -velocity  $\omega_*$  at the 1000 hPa surface, for the case of  $\alpha = 2 \times 10^3$  and  $z_0 = 1$ , using the method described in Sect. 3.1. The black contour lines are drawn at  $\pm 0.0001 \text{ Pa s}^{-1}$ . The green dotted and dashed lines indicate the phase speeds of the Lamb and Pekeris waves, respectively, calculated from the equivalent depths obtained by solving the VSE using the vertical temperature profile shown in Fig. 1

Next, we examine how the horizontal extent of the initial perturbation affects wave excitation. Figure 6 shows results for  $z_0 = 1$  and  $\alpha = 8 \times 10^3$ , where the horizontal extent is halved compared to the previous case. This leads to shorter horizontal wavelengths for both the Lamb and Pekeris waves. Notably, the Pekeris wave appears earlier, at  $t_* = 2$  h, and its node altitude matches the theoretical one by  $t_* = 4$  h.

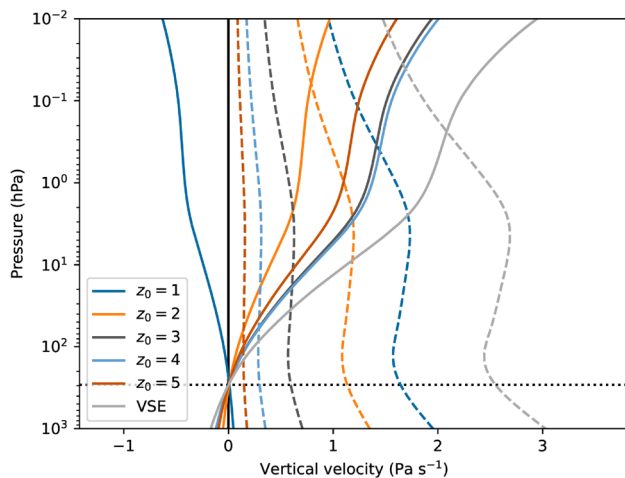
We now turn to the case where the initial temperature perturbation is imposed at  $z_0 = 3$  (around the 50 hPa level), which is above the temperature node of the Pekeris wave (located at  $z \approx 1.23$ ). As shown in Fig. 7, the horizontal sign pattern of the Lamb wave remains the same as in the  $z_0 = 1$  case, whereas that of the Pekeris wave is reversed. This phase reversal occurs because the initial perturbation is placed on the opposite side of the Pekeris mode's temperature node. Additionally, the Lamb wave amplitude decreases while the Pekeris wave amplitude increases. A comparison among Figs. 4, 6, and 7 suggests that the time required for the node altitude of the Pekeris wave to reach its theoretical value is primarily due to the spatial superposition with the leading Lamb wave.



**Fig. 6** Same as Fig. 4 except that  $\alpha = 8 \times 10^3$



**Fig. 7** Same as Fig. 4 except that  $z_0 = 3$

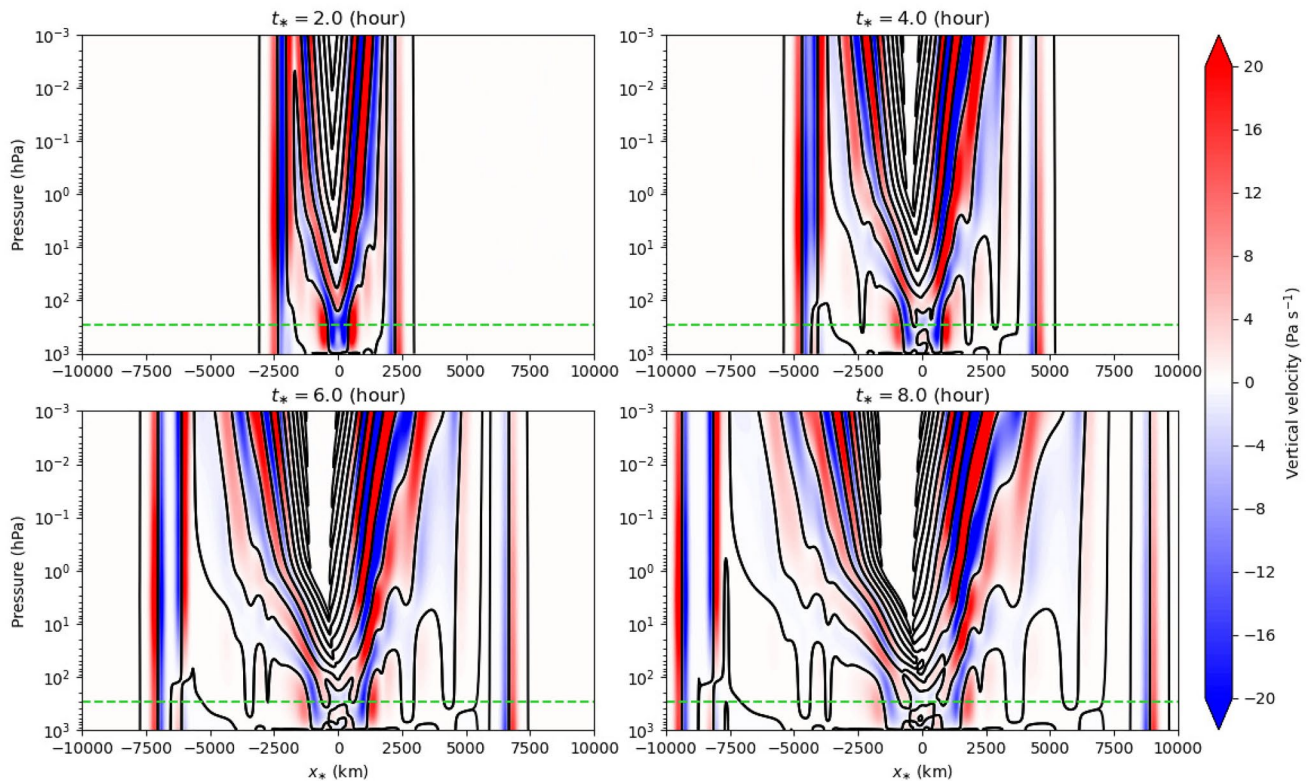


**Fig. 8** Vertical structures of the normalized  $p$ -velocity  $\omega_* \times (p_0/p_*)^{5/7}$ , associated with the leading anomalies of the Lamb and Pekeris waves (dashed and solid lines, respectively) at  $t_* = 8$  h, for the cases of  $\alpha = 2 \times 10^3$  and  $z_0 = 1, 2, 3, 4$ , and  $5$ , using the method described in Sect. 3.1. The color legend is shown in the figure. The vertical structures obtained by solving the VSE using the vertical temperature profile shown in Fig. 1 are also shown, with their amplitudes arbitrarily scaled. The black dotted line indicates the theoretical node altitude of the Pekeris wave

To clarify the dependence of the excited waves on the source altitude, Fig. 8 compares the vertical structures of the Lamb and Pekeris waves at  $t_* = 8$  h for various initial perturbation heights ( $z_0 = 1, 2, 3, 4$ , and  $5$ ). After sufficient time, the vertical structures of both modes closely match the theoretical ones, regardless of  $z_0$ . The amplitude of the Lamb wave decreases monotonically as the source height increases. In contrast, the sign of the Pekeris wave depends on whether  $z_0$  is above or below its temperature node ( $z \approx 1.23$ ), and its excitation is most efficient at  $z_0 = 4$ . These results indicate that the Lamb and Pekeris waves are most effectively excited at different altitudes, consistent with the differences in their intrinsic vertical structures.

## 4.2 Excitation of Lamb and Pekeris modes in a plane atmosphere with background winds

In this subsection, we investigate the effects of background winds on Lamb and Pekeris waves using the method described in Sect. 3.2, which assumes a plane atmosphere. Figure 9 shows horizontal–height cross sections of the normalized  $p$ -velocity at  $t_* = 2, 4, 6$ , and  $8$  h for the case of



**Fig. 9** Horizontal–height cross sections of the normalized  $p$ -velocity  $\omega_* \times (p_0/p_*)^{5/7}$  at  $t_* = 2, 4, 6$ , and  $8$  h, for the case of  $z_0 = 1$ , using the method described in Sect. 3.2. The black contour lines are drawn

at  $\pm 0.0001 \text{ Pa s}^{-1}$ . The green dashed line indicates the node altitude of the Pekeris wave obtained by solving the VSE using the vertical temperature profile shown in Fig. 1

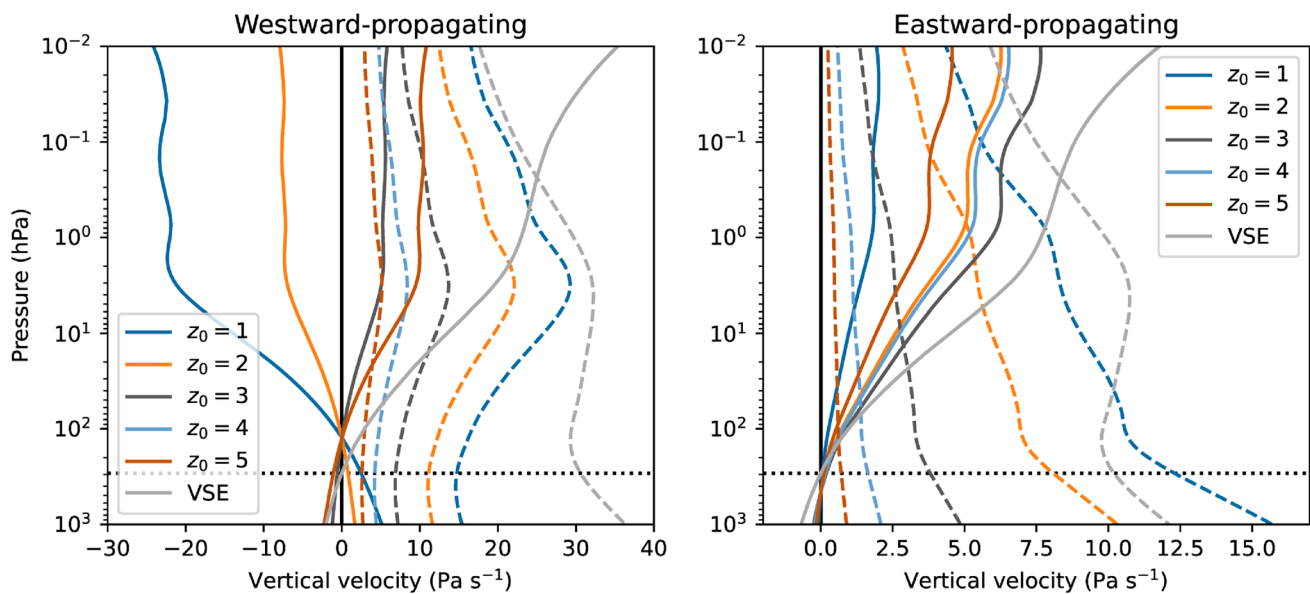
$z_0 = 1$ . After  $t_* = 4$  h, the Pekeris wave is clearly observed, and its node altitude remains nearly stationary thereafter. Notably, the node altitude of the westward-propagating Pekeris wave is higher than the theoretical value (calculated in the absence of winds), whereas that of the eastward-propagating wave is lower. At  $t_* = 8$  h, the arrival location of the Lamb wave shows little difference between the eastward and westward directions. In contrast, the Pekeris wave exhibits a significant directional asymmetry in its propagation distance. Furthermore, the eastward-propagating Pekeris wave has a smaller amplitude than its westward counterpart and displays a slight but discernible vertical phase tilt. The horizontal sign pattern also differs between the westward- and eastward-propagating Pekeris waves.

Next, we examine how the vertical structures of these waves vary with the height of the initial perturbation  $z_0$  in the presence of background winds. Figure 10 shows the vertical structures associated with the leading anomalies of the westward- and eastward-propagating Lamb and Pekeris waves at  $t_* = 8$  h for  $z_0 = 1, 2, 3, 4$ , and  $5$ . As in the case without background winds, the vertical structures of both modes are independent of the initial perturbation altitude, although they are distorted by the influence of zonal winds. Lamb waves exhibit larger amplitudes when the initial perturbation is at a lower altitude, consistent with the results in

Sect. 4.1. In contrast, the altitude of the initial perturbation that maximizes the amplitude of the Pekeris wave shifts as a result of the modification of its vertical structure and differs between the two propagation directions. Furthermore, unlike the results without background winds (Fig. 8), a sign reversal in the westward-propagating Pekeris wave is seen between the experiments for  $z_0 = 2$  and  $3$ . For the eastward-propagating wave, no sign reversal is observed for  $z_0 \geq 1$ . This difference is attributed to the fact that, in the presence of zonal winds, the node altitude for the westward-propagating Pekeris wave lies between  $z_0 = 2$  and  $3$ , while that for the eastward-propagating one has shifted to below  $z_0 = 1$ .

### 4.3 Lamb and Pekeris mode solutions under the influence of background winds

In this subsection, we present the phase speeds of Lamb and Pekeris waves calculated using the method described in Sect. 3.3. In the absence of background winds, the phase speeds are purely real, with values of  $314 \text{ m s}^{-1}$  for the Lamb wave and  $252 \text{ m s}^{-1}$  for the Pekeris wave. When the zonal wind shown in Fig. 1 is prescribed, the phase speeds of the Lamb waves show relatively small changes: the westward-propagating wave speeds up to  $326 \text{ m s}^{-1}$  ( $+12 \text{ m s}^{-1}$ ), while the eastward-propagating wave slows down to



**Fig. 10** Vertical structures of the normalized  $p$ -velocity  $\omega_* \times (p_0/p_*)^{5/7}$ , associated with the leading anomalies of westward- and eastward-propagating Lamb and Pekeris waves (dashed and solid lines, respectively) at  $t_* = 8$  h, for the cases of  $z_0=1, 2, 3, 4$ , and  $5$ , using the method described in Sect. 3.2. The color legend is shown in the figure. The

vertical structures obtained by solving the VSE using the vertical temperature profile shown in Fig. 1 are also shown, with their amplitudes arbitrarily scaled. The black dotted line indicates the theoretical node altitude of the Pekeris wave calculated without background winds

$309 \text{ m s}^{-1}$  ( $-5 \text{ m s}^{-1}$ ). In contrast, the Pekeris waves exhibit larger shifts: the westward-propagating wave has a phase speed of  $283 \text{ m s}^{-1}$  ( $+31 \text{ m s}^{-1}$ ), and the eastward-propagating one has  $217 \text{ m s}^{-1}$  ( $-35 \text{ m s}^{-1}$ ). These phase speeds are in good agreement with the results presented in Sect. 4.2. Notably, only the eastward-propagating Pekeris wave has a negative imaginary component in its phase speed, indicating a temporally decaying solution. For a zonal wavelength of  $100 \text{ km}$ , the corresponding e-folding decay time is  $0.60 \text{ h}$ .

Figure 11 shows the vertical structures of the absolute values of the log- $p$  velocity  $\hat{W}$  and the geopotential  $\hat{\phi}$  for the Lamb and Pekeris waves, along with the phases for the eastward-propagating Pekeris wave only. To clarify the influence of background winds, no normalization for the height-dependent amplitude has been applied in this figure. The amplitude of the eastward-propagating Lamb wave exhibits stronger vertical attenuation than the no-wind case, whereas the westward-propagating one shows the opposite tendency. For the eastward-propagating Pekeris wave, the amplitude remains nearly constant above the  $0.1 \text{ hPa}$  level, which, along with a slight phase tilt, indicates a wavelike structure at high altitudes.

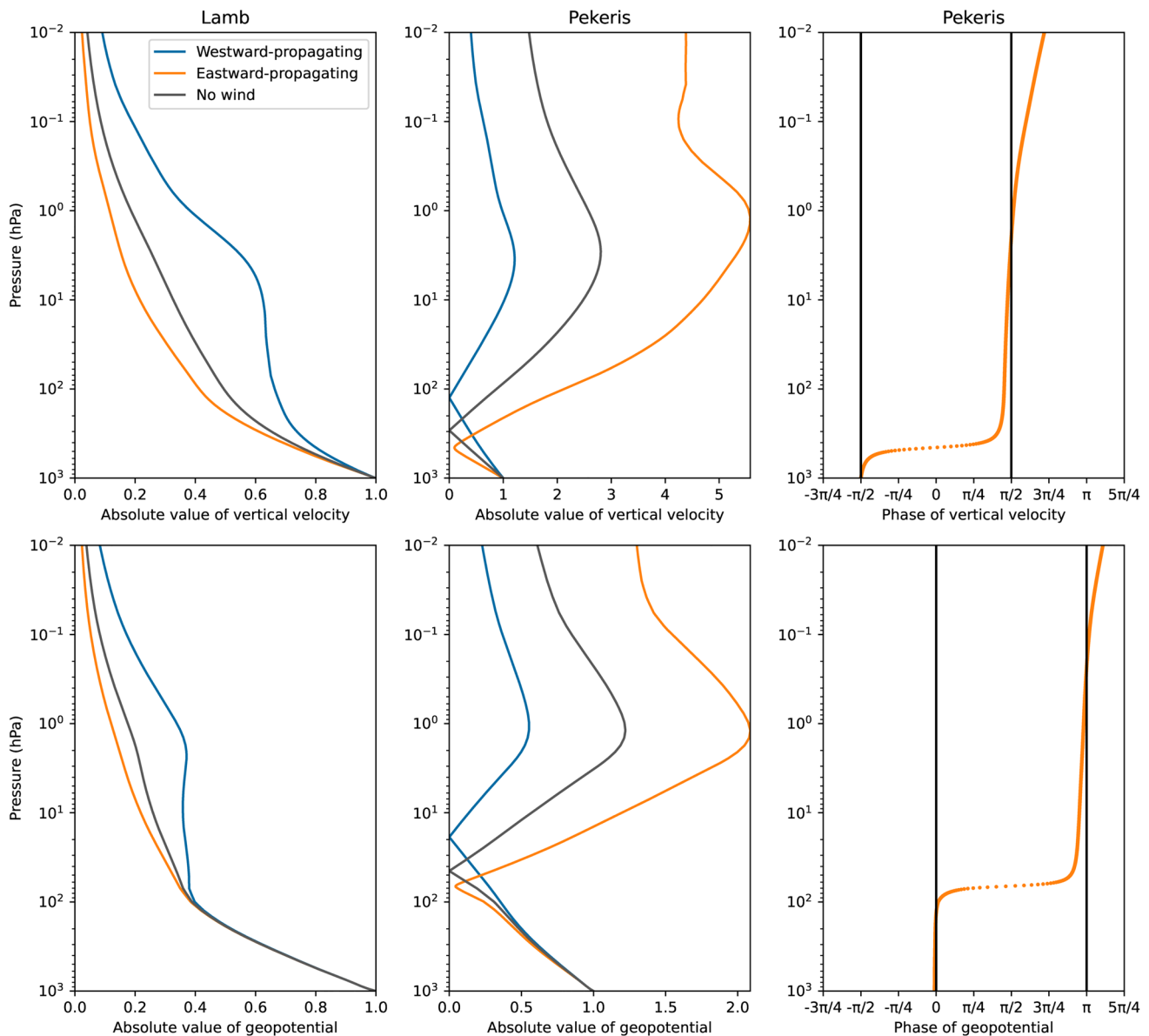
The background winds also shift the node altitudes of the Pekeris waves. For the westward-propagating wave, the nodes are located at  $126 \text{ hPa}$  for  $\hat{W}$  and  $19 \text{ hPa}$  for  $\hat{\phi}$ , which are higher than the no-wind values. In contrast, for the eastward-propagating wave, the nodes are lowered to  $457 \text{ hPa}$  for  $\hat{W}$  and  $67 \text{ hPa}$  for  $\hat{\phi}$ . These shifts are consistent with the

excitation characteristics and sign reversals observed in the numerical experiments in Sect. 4.2.

#### 4.4 Comparison of numerical results with satellite observations

In this section, using the method described in Sect. 3.4, we represent the atmospheric waves excited by the initial temperature perturbation in terms of the second derivative of radiance and compare them with satellite observations. Figure 12 shows the second derivative of radiance at  $t_* = 4, 6$ , and  $8 \text{ h}$  for  $z_0 = 1, 2, 3, 4$ , and  $5$ , calculated without background winds (Sect. 3.1). In terms of radiance signatures, consistent with the  $p$ -velocity structures in Fig. 8, the Lamb wave amplitude is larger for lower source altitudes, whereas the Pekeris wave amplitude increases for higher source altitudes. The sign of the Pekeris wave reverses depending on whether  $z_0$  is above or below its node altitude ( $z \approx 1.23$ ). Although amplitudes decrease as the waves propagate radially, the Lamb wave radiance signal remains considerably stronger than that of the Pekeris wave for  $z_0 = 1$  and  $2$ , comparable for  $z_0 = 3$  and  $4$ , and smaller for  $z_0 = 5$ . In the satellite observations (Figs. 2 and 3), the Lamb wave signal is generally slightly stronger than the Pekeris wave, and both exhibit prominent negative anomalies. This suggests that  $z_0 = 3$  provides the best overall agreement with observations in the no-wind case.

Next, we examine how the radiance signatures are modified by background winds. Figure 13 shows the results using



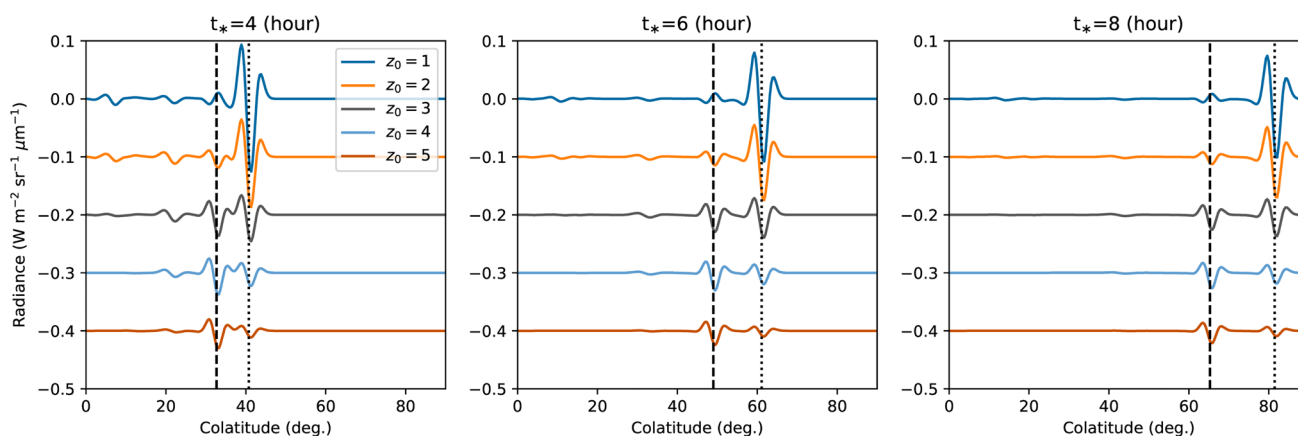
**Fig. 11** Vertical structures of the absolute values of the log- $p$  velocity  $\tilde{W}$  and the geopotential  $\tilde{\phi}$  for the westward- and eastward-propagating Lamb and Pekeris waves, both without background winds and with prescribed zonal winds. The vertical phase structures of the log- $p$

the method in Sect. 3.2. To account for the geometric attenuation on a sphere, a factor  $A/\sqrt{\sin(|x_*|/a_*)}$  is applied, where  $A$  is calibrated so that the maximum Lamb wave amplitude matches the spherical no-wind case. Unlike the no-wind case, the westward-propagating Pekeris wave reaches its maximum amplitude at  $z_0 = 1$ , and its sign reverses across its wind-elevated node altitude (see Sect. 4.3). For  $z_0 = 1, 2$ , and  $3$ , its amplitude is considerably smaller than that of the westward-propagating Lamb wave, while for  $z_0 = 4$  and  $5$ , the two are comparable. Comparing this with the westward-propagating waves in Fig. 2,  $z_0 = 4$  and  $5$  appear most

velocity and the geopotential are shown only for the eastward-propagating Pekeris wave in the rightmost panels, where additional vertical black lines are drawn to clarify the phase tilt. The color legend is shown in the figure

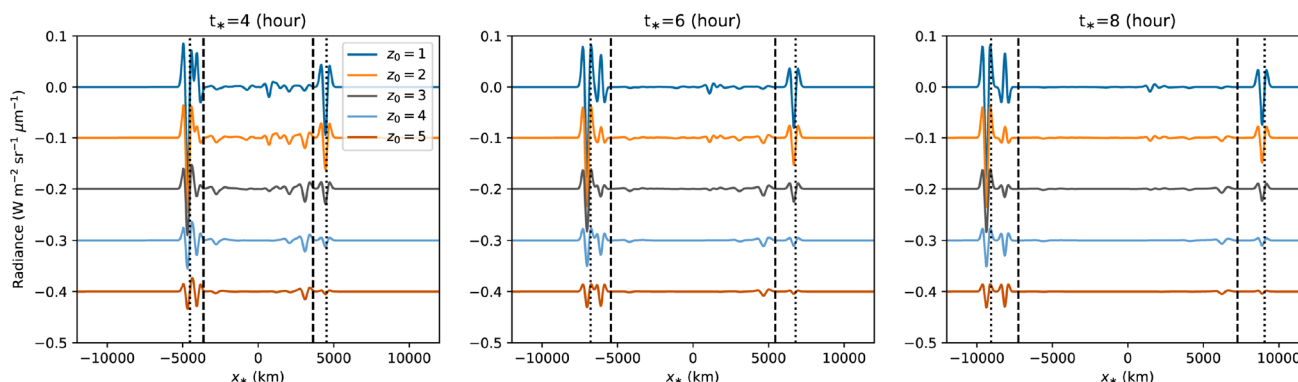
consistent with observations due to the comparable amplitudes and prominent negative anomalies.

In contrast, the eastward-propagating Pekeris wave for  $z_0 = 1$  is much weaker than for other  $z_0$  values, and its horizontal sign pattern remains consistent for all  $z_0 \geq 1$  because the node has shifted below the source region. While this mode undergoes significant wind-induced attenuation, its amplitude at  $t_* = 6$  h is much smaller than the eastward Lamb wave only for  $z_0 = 1$  and  $2$ . In the GOES observations (Fig. 3), the eastward Pekeris wave is indeed much weaker than the eastward Lamb wave, favoring  $z_0 = 1$  or  $2$ . Consequently, the optimal source altitude for westward



**Fig. 12** Second derivative of radiance at  $t_* = 4, 6$ , and  $8$  h, for the cases of  $\alpha = 2 \times 10^3$  and  $z_0 = 1, 2, 3, 4$ , and  $5$ , using the method described in Sect. 3.1. For better visibility, the amplitude is offset by  $-0.1$  for each increment of  $1$  in  $z_0$ . The black dotted and dashed lines indicate the

arrival location of the Lamb and Pekeris waves, respectively, calculated from equivalent depths obtained by solving the VSE using the vertical temperature profile shown in Fig. 1



**Fig. 13** Same as Fig. 12, except that the results are calculated including background winds using the method described in Sect. 3.2

propagation ( $z_0 = 4-5$ ) is inconsistent with that for eastward propagation ( $z_0 = 1-2$ ) under the prescribed background winds.

## 5 Summary and discussion

### 5.1 Summary

In the present study, we investigated the atmospheric response to an initial temperature perturbation representing a volcanic eruption using a linear atmospheric model. Our objective was to clarify the causes of discrepancies in the equivalent depth and the node altitude of the Pekeris wave reported in previous studies (Watanabe et al. 2022; Ishizaki et al. 2023). Specifically, we utilized the primitive equations in log-pressure coordinates on a non-rotating sphere, linearized around a basic state at rest where the temperature field depends only on altitude. An axisymmetric initial temperature perturbation was imposed at the pole, and longitudinally uniform atmospheric waves were calculated.

When an initial temperature perturbation consistent with Watanabe et al. (2022) was applied, the phase speed and vertical structure of the Pekeris wave at  $t_* = 8$  h showed good agreement with those obtained by solving the VSE following the method of Ishizaki et al. (2023). However, at  $t_* = 4$  h, the Pekeris wave was superimposed with the Lamb wave, causing the node altitude to differ from the theoretical value. When the horizontal extent of the initial perturbation was halved, the horizontal wavelengths of the excited Lamb and Pekeris waves became smaller. Consequently, the Pekeris wave at  $t_* = 4$  h behaved as an isolated wave, and its node altitude matched the theoretical value. Similarly, increasing the altitude of the initial perturbation enhanced the Pekeris wave amplitude while reducing that of the Lamb wave, leading to better agreement with the theoretical node altitude at  $t_* = 4$  h, although the waves remained partially mixed. These results demonstrate that the apparent characteristics of the excited Pekeris wave are highly sensitive to the horizontal extent and altitude of the initial perturbation.

We also examined the excitation and propagation of two-dimensional atmospheric waves on a plane using a basic

state in which both temperature and zonal wind depend on altitude. When an initial temperature perturbation consistent with Watanabe et al. (2022) was imposed, the phase speed of the westward-propagating Pekeris wave was considerably faster than in the no-wind case, while that of the eastward-propagating one was considerably slower. The node of the westward-propagating Pekeris wave was much higher than in the no-wind case, while that of the eastward-propagating one was much lower. Furthermore, the amplitude of the westward-propagating Pekeris wave was significantly larger than that of the eastward-propagating one. To clarify the underlying physical mechanisms, mode solutions for the westward- and eastward-propagating Pekeris waves were calculated using the same background fields. Their characteristics, including phase speed, node altitude, and amplitude, were consistent with the numerical simulations for both propagation directions.

## 5.2 Discussion

First, we consider the Doppler shift imposed by background winds on the Lamb and Pekeris waves. In Sect. 4.3, it was demonstrated that under the zonal wind profile shown in Fig. 1, the westward-propagating Pekeris wave is approximately  $30 \text{ m s}^{-1}$  faster than in the no-wind case, while the eastward-propagating counterpart is slower by a comparable amount. This is because the Pekeris wave has a large amplitude in the middle atmosphere (Fig. 11) and is therefore highly susceptible to the strong easterlies in that region. Compared with the Pekeris wave, the Doppler shift on the Lamb wave is smaller ( $-5 \text{ m s}^{-1}/+12 \text{ m s}^{-1}$ ), likely because its amplitude is large in the troposphere, where weak westerlies partially cancel the influence of the strong easterlies in the middle atmosphere.

Next, we discuss the discrepancy in the Pekeris wave node altitude between the simulation by Watanabe et al. (2022) and the theoretical VSE solution. As shown in Sect. 4.1, the time required for the Pekeris wave to separate from the Lamb wave depends on the initial perturbation structure and can be quite long. Since the vertical structure presented in Watanabe et al. (2022) corresponds to approximately four hours after the eruption, the wave may still have been in a transient state of spatial superposition. However, as demonstrated in Sect. 4.3, the background zonal winds at  $20^\circ\text{S}$  shift the node of  $p$ -velocity of the westward-propagating Pekeris wave to 126 hPa, significantly higher than the no-wind theoretical value (292 hPa). In the simulation of Watanabe et al. (2022), the node directly west of the HTHH is located at about 90 hPa, which is similarly significantly higher than the theoretical value. This consistency suggests that the higher node altitude reported by Watanabe et al. (2022) is primarily attributable to the influence of background winds.

The elevation of the node altitude can be qualitatively explained in the following manner. Neglecting the vertical shear, equations (29)–(31) are rewritten as follows:

$$\frac{\partial^2 \hat{W}}{\partial z^2} = -\left(\frac{N^2}{\tilde{c}^2} - \frac{1}{4}\right) \hat{W}, \quad (42)$$

$$\frac{\partial \hat{W}}{\partial z} = -\left(\frac{\tilde{T}}{\tilde{c}^2} - \frac{1}{2}\right) \hat{W} \quad (z = 0). \quad (43)$$

Letting  $\frac{N^2}{\tilde{c}^2} - \frac{1}{4} = m^2$  and  $\frac{\tilde{T}}{\tilde{c}^2} - \frac{1}{2} = \gamma$ , equations (42)–(43) are simplified as follows:

$$\frac{\partial^2 \hat{W}}{\partial z^2} = -m^2 \hat{W}, \quad (44)$$

$$\frac{\partial \hat{W}}{\partial z} = -\gamma \hat{W} \quad (z = 0), \quad (45)$$

where  $m$  represents the vertical wavenumber and  $\gamma$  represents the decay rate with height. For this westward-propagating case, by the Doppler shift due to the background wind, the phase speed  $c (< 0)$  is about  $30 \text{ m s}^{-1}$  smaller than the intrinsic phase speed  $c_0$ . Considering that  $\tilde{c} = c - \bar{U} \approx c_0 - 30 \text{ m s}^{-1} - \bar{U}$ , since there exists a strong easterly region where  $\bar{U} < -30 \text{ m s}^{-1}$  above 10 hPa, the inequality  $\tilde{c} > c_0$  holds, that is, the absolute value of  $\tilde{c}$  is smaller than the no-wind case. This implies that  $m^2$  must be larger in the strong easterly region, which corresponds to a greater curvature. In contrast,  $\gamma$  becomes smaller due to the weak winds near the surface, as can also be confirmed in Fig. 11. The change of the vertical structure due to these combined effects above and below the node results in an overall upward shift of the node altitude.

Suzuki et al. (2023) shows that the node of the meridional wind of the northwestward-propagating Pekeris wave approximately 5 h after the HTHH eruption is located at 75–60 hPa, which is somewhat lower than the theoretical one. This is considered due to the fact that, although they focus on the northwestward-propagating Pekeris wave, it reaches the northern hemisphere and is therefore influenced by the strong westerlies in the middle atmosphere. In fact, the node of the geopotential of the eastward-propagating Pekeris wave calculated in Sect. 4.3 is located at 67 hPa, which is consistent with their simulation results.

An interesting feature in Watanabe et al. (2022) is that the westward-propagating Pekeris wavefront is distinct, while the eastward-propagating one is indistinct. As shown in Sect. 4.3, the eastward-propagating Pekeris wave is modified by the background wind such that its vertical structure remains wavelike ( $m^2 > 0$ ) even in the low-temperature region near the mesopause. This allows for significant

vertical energy leakage into the thermosphere. Indeed, Fig. 9 shows that the eastward-propagating wave has a smaller amplitude and a slight vertical phase tilt in the mesosphere. Furthermore, the e-folding decay time of the eastward-propagating Pekeris wave (0.60 h for a 100 km zonal wavelength) is much shorter than the 0.32 days calculated by Ishioka (2026). While Ishioka (2026) considered energy leakage due to the high temperatures of the thermosphere, our results suggest that background winds further accelerate this decay by facilitating upward propagation through the modification of the vertical wavenumber. This rapid attenuation is also evident in satellite observations (Figs. 2 and 3). Although the present study focuses on the influence of the background wind, it is noteworthy that the amplitude of the northward-propagating Pekeris wave is smaller than that of the southward-propagating one in Fig. 2. This may be attributed to the fact that, during the winter season when the HTHH eruption occurred, temperatures near the mesopause are higher to the north of the HTHH, resulting in stronger energy leakage.

The discrepancy in equivalent depths of the Pekeris wave between Watanabe et al. (2022) and Ishizaki et al. (2023) may also arise from background wind effects. We demonstrated that several hours are required for the Pekeris wave to isolate itself from the Lamb wave, during which time it is subject to direction-dependent Doppler shifts. However, as is evident from the zonal direction results in Sect. 4.3, taking azimuthal averages as in Watanabe et al. (2022) does not necessarily ensure the complete cancellation of these Doppler effects along antipodal directions. In addition, the Doppler shift reaches up to  $35 \text{ m s}^{-1}$  in certain directions. Consequently, the Doppler shift can cause the equivalent depth estimated by Watanabe et al. (2022) (6.1 km, corresponding to a phase speed of  $244.5 \text{ m s}^{-1}$ ) to deviate from the theoretical value obtained without background winds by Ishizaki et al. (2023) (6.5 km, corresponding to  $252.4 \text{ m s}^{-1}$ ). Note that observational limitations, such as accurately estimating the eruption time and precisely identifying wavefronts against convective activity and amplitude attenuation, may also contribute to this discrepancy. Given these considerations, theoretically calculating the equivalent depth, as performed by Ishioka (2023) and Ishizaki et al. (2023), remains an essential and robust approach to understanding the fundamental properties of these waves.

Finally, we discuss the source structure estimated from the Lamb/Pekeris amplitude ratios. Without winds,  $z_0 = 3$  (50 hPa) provides the best agreement with observations. This result differs significantly from the source altitudes estimated without considering winds by Mizutani and Yomogida (2023) and Tonegawa and Fukao (2023). Such a discrepancy may arise from differences in model configurations, such as the use of a nonhydrostatic model or the

assumption of a free surface at the upper boundary in the two studies. In the presence of winds, however, the consistent  $z_0$  shifts to 4–5 for westward propagation and 1–2 for eastward propagation. These shifts occur because background winds modify the vertical structure and damping of the Pekeris wave, altering the amplitude at altitudes to which satellite bands are sensitive. Thus, accounting for wind effects is crucial for source estimation. The inconsistency in  $z_0$  between westward and eastward propagation likely stems from model approximations, such as the hydrostatic approximation, the neglect of nonlinearity, and the assumption of background fields that depend only on altitude. In addition, although the present study neglects the effect of rotation, introducing the effect would be necessary for a more quantitative discussion, given the behavior of inertia-gravity waves in the real atmosphere. Future work using full AGCMs will be necessary to investigate these processes in greater detail.

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**Author contributions** Conceptualization and supervision were performed by K. Ishioka. H. Ishizaki conducted the data analysis and numerical calculations, and was responsible for the writing of the entire manuscript. All authors have read and approved the final version of the paper.

**Data availability** For the ERA5, pressure-level data (Hersbach, et al., 2023) were downloaded from <https://cds.climate.copernicus.eu/datasets/reanalysis-era5-pressure-levels-monthly-means?tab=overview>, while model-level data (Hersbach, et al., 2017) were obtained through the Meteorological Archival and Retrieval System (MARS). Himawari-8 radiance data were downloaded from [https://www.cr.chiba-u.jp/databases/GEO/H8\\_9/FD/index\\_en\\_V20190123.html](https://www.cr.chiba-u.jp/databases/GEO/H8_9/FD/index_en_V20190123.html). GOES S-17 radiance data were obtained through the Comprehensive Large Array-data Stewardship System (CLASS). The datasets generated and analyzed in the present study are available from the corresponding author on reasonable request.

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